

FES: A modular pipeline to find, encode and select relevant facts from clinical notes

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FES: A modular pipeline to find, encode and select relevant facts from clinical notes

Abstract—x

Index Terms—information extraction, neuro-symbolic, evidence verification, natural language processing

I. INTRODUCTION

Epilepsy affects c. 1% of the UK population [1] and is characterised by recurrent, unprovoked seizures [2]. Seizure frequency is a key indicator of the patient’s disease burden and is commonly used in clinical care to determine the appropriate treatment and as a primary outcome in clinical trials. Despite the critical role of this information, the only place it is routinely recorded is in clinic letters. Clinic letters are unstructured narrative reports of a patient’s recent visit to the epilepsy clinic, and they often combine descriptions of past diagnoses, current symptoms and future treatment. This information can be used to conduct retrospective studies or to identify patient cohorts for clinical trials, but extracting it into that structured form requires experts to conduct costly, error-prone manual chart review.

Natural language processing (NLP) has been used to automatically extract key information from clinic letters. Rules-based methods successfully extracted structured facts about medications and diagnoses, but struggled with the imprecise language and diverse forms of seizure frequency statements [3]. ‘2 to 3 seizures a week’, ‘frequent bouts of seizures’ and ‘4 since her last visit’ are all valid descriptions of the same fact, but designing rules to accommodate these completely and consistently is very time-consuming and challenging. Rules-based methods are highly valued by clinicians because they produce highly interpretable results and can flexibly incorporate local domain expertise, but a different kind of tool is needed for these semantically complex tasks.

Large language models (LLMs) have emerged as a powerful tool and perform significantly better on semantically complex tasks, including seizure frequency extraction [4]. However, unlike rules-based methods the outputs of LLMs are not constrained to the exact text they were given and this presents challenges for the task of information extraction (IE). When given the specific task of estimating a patient’s seizure frequency from the provided clinic letter, an LLM occasionally provided diagnostic advice including prescriptions for the epilepsy patients. This phenomenon of ‘hallucination’ has been documented as a common error mode for LLMs and is a significant risk when used to produce clinical information at scale. Fine-tuning an LLM on the task minimises this error but requires substantial training data and computational resources.

Prior work has utilised rules as a light post-processing layer to repair malformed outputs from LLMs [5], but none examine a hybrid pipeline of LLM and rules as a cheaper alternative to fine-tuning LLMs. Rules are very effective at processing information in a transparent, logical, configurable way. LLMs are very effective at extracting rich information from semantically dense sources. Leveraging both capabilities at distinct stages can combat the concerns about LLMs being a black box that produces hallucinations, while providing greater semantic depth with significantly less time spent developing rules. A combination of LLMs and rules could produce a more transparent, flexible system.

Effective IE can be broken down into three key stages: finding relevant candidate facts, encoding those facts into a particular form, and selecting which facts to keep and which to filter out. A clinic letter can describe multiple seizure events with varying degrees of precision on different occasions. Frequencies can be expressed as a rate, a vague quantity or a duration. Deciding which of those facts to preserve and how to represent them is a subjective clinical policy. The same evidence can be interpreted differently by different clinicians or presented in multiple forms for distinct use cases. A flexible system would recognise this key feature of clinical information and make information available for reuse across multiple scenarios. A transparent system would allow users to see the final answer and the steps taken to determine it.

This paper evaluates the performance of LLMs and rules across those three stages: find, encode and select. It was evaluated on a set of 1,200 epilepsy clinic letters with a single seizure frequency label per letter annotated by experts. A direct comparison to Gan *et. al* [3] demonstrates that a modular system combining LLMs and rules can deliver accuracy comparable to a fine-tuned LLM while preserving a richer chain of evidence. The system is available as an open source package here along with a web app demonstrating key features of the pipeline here .

II. LITERATURE REVIEW

The best approach to extract seizure frequency information from epilepsy clinic letters is an active area of research, with experiments using rules-based methods [4], [6], transformers-based methods [3], [5], [7], [8] and a combination of both [9], [10] all published this decade. Rules-based methods struggled with the lexical variety of seizure frequency descriptions. A key challenge with both rules-based methods and BERT-based methods was their failure to generalise to letters from a different hospital. [6], [10]. LLMs were the first method to

demonstrate strong generalisation to a new dataset [3], but this was achieved by fine-tuning a model on over 1,000 synthetic letters. This paper explores whether similar performance can be achieved without any additional training or fine-tuning.

Rules have been used as a form of minimal post-processing to repair malformed outputs from LLMs [5], but none of the research has explored whether a hybrid pipeline combining an LLM and rules at distinct stages can achieve superior performance than either component alone. Rules were used to normalise frequencies and dates as a separate layer in two BERT-based pipelines [9], [10], but this has not been replicated yet for modern LLMs. Both BERT and LLMs are based on the same underlying transformers architecture, but LLMs are distinguished primarily by the fact they are trained on massive datasets with massive parameter counts which led to the emergent capability of in-context learning [11]. This capability has been shown to transfer to clinical information extraction among many other domains, and it has been shown that giving the LLM clear instructions and examples can replace the need for complex post-processing [12]. This paper explores whether LLM instructions and examples can be combined with complex post-processing to replace the need for fine-tuning.

Clinical IE has been defined as the automatic extraction and encoding of clinical information from text [13], composed of the two distinct tasks named entity recognition (NER) and concept encoding [14]. Despite this, the seizure frequency extraction pipelines using LLMs bundle both tasks into a single step by asking the model to output a normalised label e.g. '4 per month' from a raw statement 'about once a week'. A third task exists in many clinical extraction pipelines but goes unmentioned: selection. This was a distinct stage in prior rules-based systems [15] but is bundled into a single LLM prompt with the two other tasks. In a clinic letter with multiple seizure frequency statements, if the task is to output a single label describing the patient's current seizure frequency burden, the task contains three components. The model needs to find relevant seizure frequency statements in the text, encode them into a normalised form and select the one that best answers the question. This paper explores whether defining find, encode and select as separate stages in a modular pipeline can deliver comparable performance while preserving more of the evidence for downstream use.

Academic papers tend to focus exclusively on benchmark accuracy while practitioners tend to value transparency, configurability and efficiency at the cost of some accuracy [15]. Model interpretability is also a critical factor clinical settings [14]. Clinicians are trained to use deductive logic to draw conclusions and they expect systems to expose transparent, interpretable evidence that they can reliably use to make decisions that demand high accountability. Past work utilises evidence spans as a training input [3] for fine-tuning or as data exhaust as part of using a rule-based methodb4. This paper explores whether preserving a transparent evidence chain from exact text to final answer can make the output both more interpretable and more accurate.

Thank you for asking me to review your patient. I am concerned that her epilepsy has worsened in recent months, with a clear pattern of deterioration. She and her partner report increasing clusters of events correlating with the perimenstrual period, suggesting a catamenial component. In particular, she describes brief focal episodes with preserved awareness that tend to occur as she is settling to sleep, at the point of drifting off, now happening on most nights of the week. These events are accompanied by a rising epigastric sensation, facial pallor, and right-hand automatisms lasting 30–60 seconds, followed by fatigue and headache. In addition, she has had three generalised tonic–clonic seizures in the last six weeks, two of which occurred during the same perimenstrual window.

Fig. 1: An example clinic letter with two distinct seizure frequency statements highlighted as potential candidates.

TABLE I: Dataset summary

		Development	Test
Number of letters		750	450
Seizure frequency	Frequent	51%	50%
	Infrequent	17%	18%
	Unknown	17%	17%
	Seizure free	15%	15%

III. METHODS

A. Data

The dataset is a collection of epilepsy clinic letters which record a UK patient's consultation with an epilepsy specialist, often a neurologist or specialist nurse [3]. The NHS requires that clinic letters are sent to the patient's GP after all outpatient visits such as these. They are often the only record of a patient's diagnoses, treatments and outcomes in neurological disorders, and they are written in narrative form. An example clinic letter is shown in Figure 1.

Gan *et al.* [3] produced this dataset of synthetic letters modelled on real clinic letters from King's College London hospital. The real letters were used to create 10 template letters and they were combined with 500 short descriptions of seizure frequency statements produced by GPT-5 to create the full set of 15,099 letters. A sample of 1,200 letters was used in this paper. Each seizure frequency statement was annotated by trained experts to create the gold-standard labels. Each letter has a single label representing the patient's current seizure frequency burden.

Minimal preprocessing was applied to the raw data. Letters and annotations were stored in a single .json file. Letters are minimally cleaned to correctly parse quotes, new lines, tabs, the $\leq$ sign and nulls. The raw letters were ingested in the pipeline in full; no tokenisation, section-splitting or truncation was applied.

The dataset was split into a development set of 750 letters and a test set of 450 letters, stratified by their seizure frequency category (frequent, infrequent, unknown, seizure free) to bal-

TABLE II: Method configuration by pipeline stage

Method	Find	Encode	Select
Rules only	Rules	Rules	Rules
LLM and Rules	LLM	LLM and Rules	Rules
LLM only	LLM*	LLM	LLM

*A single LLM prompt is used for the find and encode stages. A second LLM prompt is used for select.

ance the distribution across both splits. The exact breakdown can be seen in Table I.

B. Architecture

The pipeline is divided into three key stages: find, encode, select. First, the pipeline finds seizure frequency events that fit the description in the raw text. Next, the pipeline takes the raw text from those candidates and encodes them into a normalised form. Finally, the pipeline selects the final answer from the set of candidate facts, either combining them together or choosing one. Figure 2 illustrates the architecture of the pipeline with a real example from the development set.

This paper evaluates three main methods on this pipeline: Rules only, LLM only, and LLM and Rules. Table II summarises their configuration at each stage. Rules only and LLM only serve as the baselines to illustrate how effectively each method performs in isolation. LLM and Rules illustrates the ideal combination of the two: the LLM finds and encodes relevant seizure frequency events, then rules are layered on top to encode and select the final answer. The LLM prompt design and rule design sections below describe the implementation details. The combined LLM and Rules method could be configured in multiple other ways, but this is the optimal combination. Further configurations are described and evaluated in the results sections.

C. LLM Prompt Design

The most consequential component of the LLM and Rules and LLM only methods is the single prompt used for the find and encode stages. It is composed of three main elements:

- 1) Instructions to identify relevant seizure frequency facts at the appropriate granularity
- 2) Examples of how the seizure frequency facts should be represented
- 3) The schema to represent them in

A key design choice in this pipeline is to represent multiple seizure frequency events with exact evidence to enable a subsequent LLM call or rules to review that evidence and refine the final answer. The model finds multiple candidate events and selects the best candidate, but that answer can be updated.

Key instructions include:

- 1) When several current seizure types are present, select the highest current or recent burden
- 2) When the letter gives an overall current count and a breakdown by type, select the overall count

TABLE III: Find event categories

Category	Use
frequency_rate	A countable or range rate
cluster_frequency	How often clusters occur
seizure_free	A seizure-free statement
last_event_only	A dated last event without a rate
unknown_frequency	Seizures discussed, no usable rate
no_reference	No usable frequency evidence

TABLE IV: Seizure-frequency label forms

Family	Written forms
Rate	N per unit; N per N unit; N per N to N unit; N to N per unit; N to N per N unit; multiple per unit; multiple per N unit; N per multiple unit
Cluster	cluster per unit, N per cluster; cluster per unit, range per cluster; cluster per unit, multiple per cluster; unknown cluster count
Seizure free	seizure free for a duration; seizure free for a vague duration
Sentinel	unknown; no seizure frequency reference

- 3) Use no_reference only when the letter has no usable seizure frequency evidence, and use unknown_frequency when seizures are discussed but frequency is unclear
- 4) Keep seizure-free statements separate from unknown or last-event statements, and do not select seizure-free if other current seizure-like events remain active

The model is instructed to assign each event to one of the six categories described in Table III and assign a label to each event in one of the forms represented in (Table IV).

D. Rule Design

E. Pipeline Optimisation

The development set was used to optimise the two key elements of the pipeline: the LLM prompt instructions and the rules. The focus of the optimisation process was to improve accuracy while keeping the instructions and rules simple and generalisable. Qualitative error analysis was conducted to diagnose the cause of poor performance in particular areas and identify common error modes. Hypotheses were developed and tested to tune the prompt and rules iteratively.

Through iterative error analysis on both the LLM and the rules, it became apparent that the complexity of the task was often a function of competing rules and guidelines. Often, improving a rule or a prompt instruction to address one class of error would harm another area because annotation policies are complex. This complexity is a challenge for human annotators also; Fonferko-Shadrach *et al.* described seizure frequency as the most difficult category to annotate and their inter-annotator agreement of 0.47 shows annotators often disagreed about how to represent seizure frequency statements.

Decomposing the task into three stages allowed for cleaner separation of the policies on which types of facts to extract and which criteria to apply to exclude specific variants of them. Decomposing the task into three stages also made it possible to

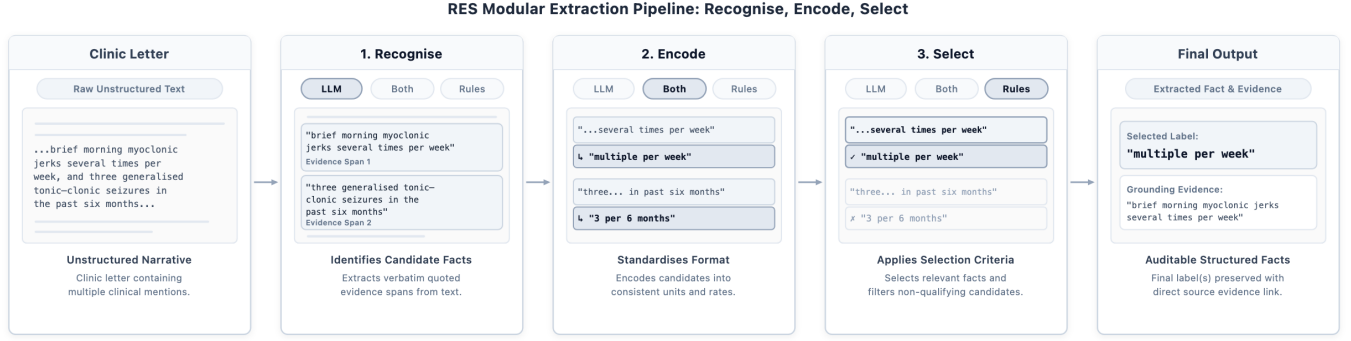


Fig. 2: A clinic letter passes through three stages in the pipeline - find, encode and select - to generate structured labels with the exact text used from the letter to produce the label. Each of the three stages can be configured to use an LLM, rules or both. The example shown is the task of extracting a single label representing the patient’s current seizure frequency burden.

evaluate the performance at each individual stage and identify the bottlenecks in the system. It is this combination of a cleaner task decomposition for the pipeline and a clearer view into the pipeline that enabled steady, incremental gains to materialise from the optimisation process.

F. Evaluation

To enable a direct comparison with the results in Gan *et. al*, the pipeline is evaluated using the same micro-F1 score. They calculate the F1 on two categories; a fine-grained purist score and a coarser ‘pragmatic’ score. The more precise purist F1 is the primary metric in this paper, but pragmatic F1 is used for frequency bands with very small sample sizes (<20). A full specification of the categories is in Table V. The rationale for categorising labels is so that semantically identical phrases are matched when exact labels differ; so that daily, once per day and every day would be considered correct matches to the gold label ‘1 per day’. This evaluates the model on how effectively it can identify the correct frequency rather than replicate precise phrasing. Further explanation on the categories can be found in the original paper from Holgate *et. al*.

The pipeline is evaluated on a held-out test set of 450 letters. These letters were never used to tune prompts, rules or any other aspect of the pipeline. The test set is therefore used to evaluate how effectively the model generalises to unseen data. These results are compared against the evaluation set labelled Synthetic (1,166) in Gan *et. al* which was also a held-out sample from the synthetic dataset, distinct from the training set used to fine-tune the models. It is important to note that Ganet. *al* also tested their results a distinct set of real clinic letters to demonstrate generalisation to real data; our results and comparisons are limited purely to the synthetic data.

The primary objective of this research is to identify which component performs best at which stage. Table II shows the 3 methods used in the evaluation: rules at every stage, the LLM at every stage or a combination of LLM and Rules. Each method was evaluated at each stage: find, encode and select.

TABLE V: Seizure frequency categories: Purist and Pragmatic

Purist	Pragmatic
Daily	Frequent
More than weekly, less than daily	
Once a week	
More than monthly, less than weekly	
Once a month	Infrequent
More than every 6 months, less than monthly	
Once every 6 months	
Less than once every 6 months	
Unknown	Unknown
Seizure free	Seizure free

This produces a 3x3 matrix of results that answer two key questions:

- 1) Among the three methods, which performs best: rules, LLM, or a combination of both?
- 2) How much incremental value is there from the additional encode and select stages?

The primary hypothesis is that a combination of an LLM plus rules will perform better than either on their own, because they have complementary strengths and those strengths are particularly well suited to distinct stages in the pipeline. The secondary hypothesis is that multiple discrete stages will improve performance for the LLM, regardless of whether the subsequent stages are rules or additional LLM calls. This builds on the idea that task decomposition is an effective form of context engineering, and is particularly useful in scenarios where the prompt is long and contains many layers of instructions.

IV. RESULTS

A. Benchmark Performance

FIRST PARAGRAPH NEEDS TO SUMMARISE THE APPROACH TO THE PROBLEM (DATA, EVALUATION) AND NOVEL METHOD CONTRIBUTIONS

The LLM and Rules method achieves state-of-the-art results (SOTA) for extracting seizure frequency from epilepsy

TABLE VI: Comparisons to past seizure frequency papers

Author	Year	Method	F1 score
Xie et al.	2022	Fine-tuned BERT	0.83*
Decker et al.	2022	Rules	0.82**
Fonferko-Shadrach et al.	2024	Rules	0.66
Fernandes et al.	2024	Fine-tuned BERT	0.37
Holgate et al.	2024	Prompted LLM	0.73
Holgate et al.	2025	Fine-tuned LLM	0.68
Gan et al.	2026	Fine-tuned LLM	0.81
Brown & Gan	2026	LLM and rules	0.86

*Drops to 0.73 on texts with a frequency statement.

**Drops to 0.4 on the external test set.

TABLE VII: LLM and Rules: Purist and Pragmatic micro-F1

Purist	<i>n</i>	Purist F1	Pragmatic F1
Daily	41	0.92	Frequent: 0.95
More than weekly, less than daily	123	0.91	
Once a week	5	0.89	
More than monthly, less than weekly	58	0.90	
Once a month	18	0.94	Infrequent: 0.82
More than every 6 months, less than monthly	55	0.81	
Once every 6 months	1	0.00	
Less than once every 6 months	6	0.60	
Unknown	76	0.74	Unknown: 0.74
Seizure free	67	0.90	Seizure free: 0.90

Rows with $n < 10$ shown in grey.

letters. It achieved a purist micro-F1 score of 0.86, exceeding the prior benchmark performance of 0.81 in Gan *et al.* set by a model that was fine-tuned and evaluated using this same dataset of synthetic letters. While that is the only paper to evaluate the same dataset with the same metric, Table VI includes a summary of all 7 papers that reported a seizure frequency extraction score this decade.

This method is particularly effective at identifying countable rates, including high-frequency events. A common theme in previous papers is that their method performed best when the task was predicting ‘unknown’ or no seizures. Xie et. al reported an F1 score of 0.92 on texts that contained no seizure frequency statement and 0.73 when there was a seizure frequency reported. Gan et al. and Holgate et. al both report unknown as their highest scoring category at 0.89 and 0.87 respectively, while daily (0.7, 0.41) and more than weekly (0.68, 0.55) are much lower. In this method, daily (0.92) and more than weekly (0.91) are among the best performing categories, significantly above unknown (0.74). The full performance breakdown across purist and pragmatic categories can be seen in Table VII.

B. Method Performance by Stage

The LLM and Rules method performs better than the LLM or rules alone at every stage: find, encode, select (Table VIII). The LLM is significantly better at the initial find stage vs. rules alone (0.79 vs 0.65). The LLM and Rules

TABLE VIII: Purist micro-F1 at find, encode, and select

Method	Find	Encode	Select
Rules only	0.65	0.65	0.72
LLM + rules	0.79	0.80	0.86
LLM only		0.79*	0.79

*A single LLM prompt is used for the find and encode stages.

combination then adds incremental gains at the encode (+0.01) and select (+0.06) stages, while the subsequent LLM stages make a very marginal difference (+0.006). This illustrates that the LLM and rules have complementary strengths that are particularly effective at discrete stages.

The LLM is particularly strong at the find stage: identifying high quality candidates in the text and providing an initial proposed answer in the required form. This initial stage is equivalent to the entire extraction pipeline in most prior papers: find, encode and select all at once. The LLM at this stage performs at a level comparable to the best prior methods, achieving performance in a single prompt that previously required fine-tuning.

Rules are particularly effective at the select stage: reviewing all of the candidates proposed by the model and determining which should be the final answer. This can mean either choosing a different candidate, such as choosing a frequency candidate when the model has chosen unknown, or performing a mathematical calculation on the selected evidence to update the answer, such as counting the total events over a monthly seizure diary. This is a direct consequent of the pipeline design. The find phase outputs multiple candidate events with evidence quoted directly from the text, rather than just a single flat answer. Rules leverage that richer evidence to deliver significant incremental gains.

C. Model Comparison

The LLM and Rules method shows consistent, incremental performance gains on each of the six models evaluated (Figure 4). The average performance gain was +0.07 from find to select, with Luna showing the largest gain of +0.10. Model performance on purist F1 closely correlates to overall medical intelligence as measured by Artificial Analysis’ Healthcare and Medical Index. They cluster into three groups: strong (Gemini 0.86, Grok 0.85), medium (DeepSeek 0.82, Luna 0.79), weak (Qwen 0.76, Gemma 0.72) which largely reflect the model size and cost.

Local models that fit on consumer-grade hardware now deliver superior performance to models that required eight Nvidia V100 GPUs two years ago (Holgate, 2024). Qwen’s performance of 0.76 was achieved running locally on a laptop with a single GPU, surpassing Llama 2’s performance of 0.73. That is a critical development as it means hospitals can use their own servers to securely process patient data to reliably extract clinical information like this. DeepSeek is small enough for many hospital servers to host and it performs better than a model fine-tuned on 1,797 synthetic letters.

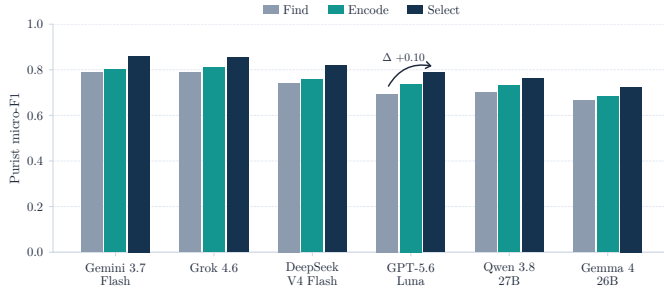


Fig. 3: LLM and rules. Every model sees consistent performance gains in the encode and select stages. Luna saw the biggest performance gain (0.69→0.79).

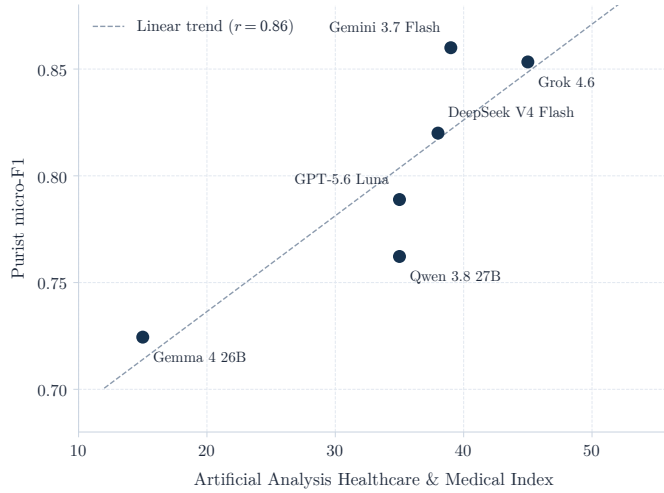


Fig. 4: The smartest models are the best frequency extractors.

D. Model Configuration

Raising the reasoning effort level did not improve performance. Gemini’s purist-F1 score was 0.86 at low and 0.84 at high; a paired difference of +0.02, 95% CI -0.01 to 0.04; $p = 0.31$. Extra reasoning adds cost and latency without improving extraction.

The temperature setting also had no significant effect on the results. The F1 score was 0.86 on a temperature of 0 versus 0.84 at a temperature of 1 (paired difference +0.02, 95% CI -0.01 to 0.04; $p = 0.20$). A temperature of 0 was an appropriate default across models and Luna at a temperature of 1 is a valid comparison.

The task is tightly defined with explicit schema and output requirements, which limits the expected variability by design. This partly explains why neither setting had a significant effect on results and it indicates the robustness of the method.

E. Development vs. Test Results

Rules only perform better on the development set but significantly worse on the test set (Figure 5), which repeats the pattern of overfitting commonly found in the literature. Rules are brittle and tend to overfit specific patterns in the training data, generalising poorly to unseen data. The Rules

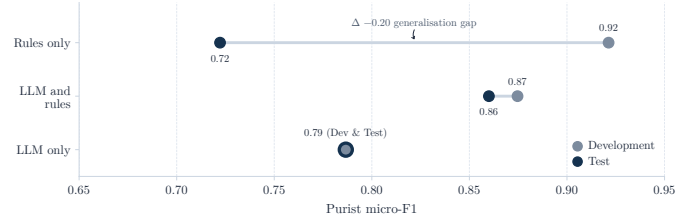


Fig. 5: Rules only performs best on the development set but overfit. LLM only generalises best with consistent performance. LLM and rules combined produce the best results overall, even with a generalisation gap of -0.01 (0.87→0.86).

only method achieves an F1 score of 0.92 on the development set, but performance declines significantly (-0.20) on the test set. LLM only achieves a consistent score of 0.79 across both splits which shows the method generalises well. The LLM and Rules method combines the best parts of each: it performs significantly better on the development set than LLM only (+0.08) and it has a significantly smaller generalisation gap than rules only (-0.01). The LLM does the heavy lifting of matching patterns across the entire text, allowing the rules to focus on generalisable patterns in the refined evidence set.

F. Error Analysis

The worst performing category was unknown with an F1 score of 0.74. 69% of the errors (37/54) are incorrect selections of unknown. The full error distribution is shown in the confusion matrix (Figure 6).

Illustrative examples of the three most common errors - infrequent → unknown (16), frequent → unknown (13), and seizure free → unknown (8) - are shown in Figure ???. The errors reflect a cautious approach when the evidence is ambiguous, for example the model recognised the phrase ‘these episodes crop up most shifts’ as the primary seizure frequency description but refused to make an assumption about how many shifts the person has. This cautious assessment is justifiable and a key benefit of this method is the exact evidence is preserved, so any downstream users can see the proposed answer while also drawing their own conclusion from the neatly packaged evidence.

G. Development Environment

Experiments using local language models took place on a Dell XPS 16 9640 running Windows 11, with an Intel Core Ultra 9 185H, 32 GB RAM, and an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB VRAM). Ollama v0.32.15 was used to serve the models. Two models were run on local hardware (Qwen 3.8 27b, Gemma 4 26B).

Experiments using hosted APIs were run from a Mac mini M1 with 16 GB unified memory on macOS 26.5.2. API calls were made directly to the provider (OpenAI, DeepSeek) or through model routers (OpenRouter, AI Gateway) Four hosted models were used (Gemini 3.7 Flash, GPT 5.6 Luna, Grok 4.6, DeepSeek v4 Flash).



Fig. 6: Unknown was the category with the most incorrect predictions from all 3 categories.

All models used a temperature of 0 except Luna, which only accepts a temperature of 1. An ablation was run on Gemini using a temperature of 1. All models used a thinking or effort level of 'low'. Ablations were run on Gemini using thinking levels of medium and high.

Python 3.14.4 was used to write the program. DSPy 3.2.1, Pydantic 2.13.4 and LiteLLM 1.87.1 were utilised as core packages. DSPy was used to co-ordinate the model calls and require structured JSON output. Pydantic was used to validate the structured output conforms to the specified schema. LiteLLM was used to deliver API calls to Ollama and the model routers.

V. DISCUSSION

This paper shows that a combination of an LLM and rules can extract an epilepsy patient's current seizure frequency burden as accurately as a fine-tuned LLM [2]. An LLM small enough to fit on a hospital's local servers can securely and reliably extract this information with no additional training or fine-tuning; a single LLM prompt designed to extract rich evidence followed by some rules to refine that evidence is just as accurate. This method requires significantly less computing power, data and expertise than training or fine-tuning an LLM, lowering barriers to adoption. Broader use of automated clinical extraction pipelines can help expand the scope of real-world evidence (RWE) as an alternative to expensive clinical trials [2].

The method is particularly effective at identifying patients with the greatest disease burden; those with the most frequent seizures. F1 scores of 0.92 for daily and 0.91 for more than weekly are substantially (0.2) higher than prior methods [2]. This could be used in a few scenarios. It could be used in clinical care to flag patients with a seizure burden above a typical threshold for the clinician's attention. It could be used in cohort selection to identify patients to be enrolled in clinical

trials for a new medication or to be recommended for surgery. It could be used to build a phenotypic profile of patients to be used in retrospective analyses to understand how these patients differ from the average. Finally, it could be used as an outcome variable in predictive models to identify patients likely to experience frequent seizures and recommend an early intervention [2].

The latest generation of LLMs are now capable enough to reliably populate rich schemas of facts and relations. Past work in this area has typically tasked the LLM with producing a single answer in a fully normalised form [2]. This fails to leverage the rich semantic depth both in clinical narratives and within the model's capabilities. Clinicians value unstructured narrative because it is more expressive; it allows them to convey their reasoning, uncertainty and vague impressions. These nuanced observations are not only more honest representations, they are more accurate and reliable [2]. LLMs are capable of preserving and reasoning over those nuances, and preserving this richer evidence chain can help build a more trustworthy, transparent system. In a clinical setting, a more explainable system is often a more useful system [2].

There are a few limitations. Synthetic data was used for all evaluations, and while past research has shown models fine-tuned on this dataset generalise effectively to real world patient data [2], the methods tested in this paper remain untested on real patient data. Past research has shown that rules tuned on one corpus can generalise poorly to other corpuses, so the gains seen from rules in this method may not generalise to other datasets. As this method was applied to a narrow task of seizure frequency extraction, future studies should evaluate whether it generalises to other common tasks in clinical IE. Alternatively, the modular pipeline applied to this seizure frequency task could be reconfigured to an agentic pipeline, with LLMs deciding how to encode and select facts themselves in a more general framework. This kind of pipeline has shown promising results on clinical notes in oncology [2] and cardiology [2].

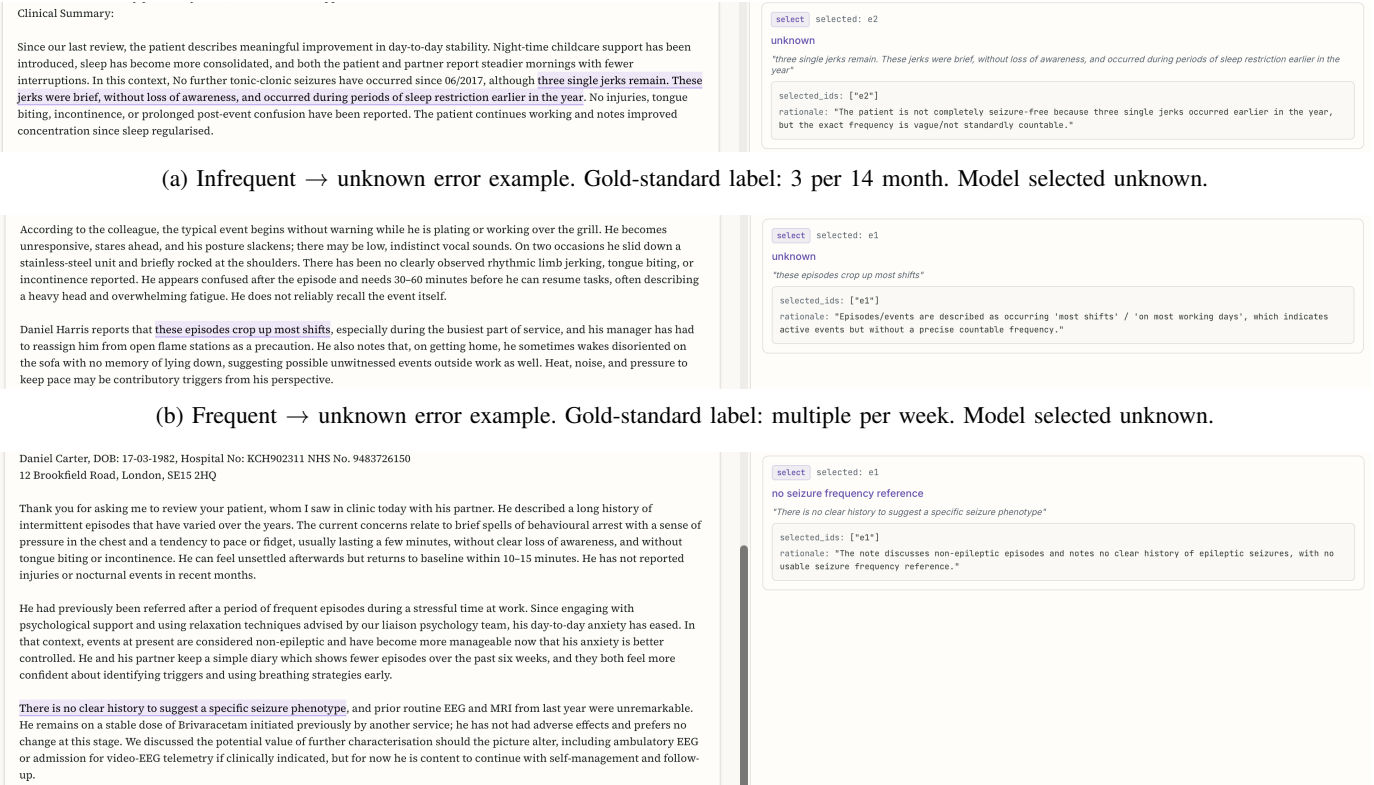
VI. CONCLUSION

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Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2].

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(c) Seizure free → unknown error example. Gold-standard label: seizure free for multiple month. Model selected no seizure frequency reference (unknown).

Fig. 7: These examples illustrate the model choosing unknown when the gold-standard label was in the infrequent (a), frequent (b) and seizure free (c) categories. The rationale illustrates how the model interprets the ambiguous evidence and why it chooses unknown.

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